

Then, we have to estimate the following five parameters using actual trade data:

$$a_1, a_2, b_1, b_2, \gamma$$

In practice, it is often difficult to find statistically significant robust and stable parameters over time. Often parameters jump around from period to period and from stock to stock. Furthermore, these parameters frequently take on counterintuitive values such as the case if either $a_2 < 0$ or $b_2 < 0$ which would imply cheaper costs as we increase the quantity of shares traded. This would also create an arbitrage opportunity. For example, an investor would be able to purchase a large number of shares of stock and then sell smaller pieces of the order at higher prices. While this may be appropriate for a large bulk purchase at an outlet store it does not hold true for stock trading.

The Almgren & Chriss model is:

$$P_k = P_0 + \sum_{j=1}^k f(x_j) \cdot e^{-\gamma \cdot (k-j)} + \sum_{j=1}^k g(x_j) + \sum_{j=1}^k \varepsilon_j \quad (4.2)$$

I-STAR MARKET IMPACT MODEL

This section provides an overview of the I-Star market impact model. The model was originally developed by Kissell and Malamut (1998) and has been described in *Optimal Trading Strategies* (Kissell and Glantz, 2003), *A Practical Framework for Estimating Transaction Costs and Developing Optimal Trading Strategies to Achieve Best Execution*, (Kissell, Glantz, and Malamut (2004)), and *Algorithmic Trading Strategies* (Kissell, 2006). The model has greatly evolved since its inception in order to accommodate the rapidly changing market environment such as algorithmic trading, Reg-NMS, decimalization, dark pools, defragmentation, and a proliferation of trading venues, etc. A full derivation of the model is provided below with additional insight into where the model has evolved to incorporate industry and market microstructure evolution.

The I-Star impact model is:

$$I_{bp}^* = a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \quad (4.3)$$

$$MI_{bp} = b_1 \cdot I^* \cdot POV^{a_4} + (1 - b_1) \cdot I^* \quad (4.4)$$

$$TR = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{S}{ADV} \cdot \frac{1 - POV}{POV}} \cdot 10_{bp}^4 \quad (4.5)$$